

TANISHA RAJ

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EDUCATION

- National Institute of Technology , Durgapur** 2023-2027
Bachelor's In Technology - Metallurgical and Materials Engineering Durgapur, India
 - CGPA: 8.58/10.00
- Hem Sheela Model School** 2023
(10+2) AISSCE'23 Durgapur, India
 - Percentage: 94.2%
- Hem Sheela Model School** 2021
AISSE'21 (Class:10) Durgapur, West Bengal
 - Percentage: 98.2%

WORK EXPERIENCE

Summer Undergraduate Research Graduate Excellence (SURGE) May 2026 - July 2026

Indian Institute of Technology, Kanpur, Uttar Pradesh



- Selected for the SURGE 2026 Research Internship at IIT Kanpur under the guidance of Prof. Kallol Mondal.
- Conducted electrochemical corrosion studies using OCP, EIS, LPR, and Cyclic Polarization.
- Performed material characterization using W-SEM, FTIR, Raman, and UV-Visible Spectroscopy.
- Analyzed corrosion behaviour and adsorption mechanisms of organic inhibitors for high-chromium steel.

Summer Internship

June 2025 - July 2025

SPS Steels Rolling Mills LTD. (Shakambari Group), Durgapur, West Bengal



- Gained hands-on experience across the complete steel manufacturing process, including DRI, Steel Melting Shop (SMS), Rolling Mill, and Structural Mill operations.
- Analyzed DRI production processes, focusing on rotary kiln operation, heat transfer, and solid flow dynamics.
- Studied steelmaking and continuous casting processes, including melting, alloying, and MS billet production in the Steel Melting Shop.
- Developed understanding of hot rolling, Thermex quenching, roll pass design, deformation mechanics, quality control, and industrial safety practices.

SKILLS

- Materials Characterization:** W-SEM, Optical Microscopy, FTIR, Raman Spectroscopy, UV-Visible Spectroscopy
- Metallurgical Software Tools:** VersaStudio, GNU Octave (DFT Calculations), OriginPro, Microsoft Excel
- Soft Skills:** Leadership, Time Management, Teamwork, Public Speaking
- Technical Skills:** C++ ,Python, Machine Learning

PROJECTS

Project A: Predicting Yield Strength of Steel Alloys by Regression January 2025

Tools: [Python , Numpy, Matplotlib, Scikit-learn]



- Developed a **regression-based model** to predict **steel properties** and **analyze material behaviour**
- Applied data preprocessing and optimization techniques for reliable predictions
- Visualized property relationships** to **gain insights into strength-ductility trends**

POSITION OF RESPONSIBILITY

- Convenor** February 2024 - Present
ISTE Student's Chapter NIT Durgapur
- Event Management Head** February 2025 - Present
IIM Student's Chapter NIT Durgapur

ACHIEVEMENTS

- Selected in prestigious SURGE Programme 2026, IIT Kanpur, under Fully Institute Funded Category**
(Top 60 nationwide)
- Secured Third position in school in AISSE'21 (Class 10)**
Hem Sheela Model School